Administrarea indecsilor
(Cap. 12)

1) Crearea unui index de tip B-Tree:

```
SQL>  create index scott.emp_name_idx
          on scott.emp(ename)
          pctfree 30
          storage(initial 200k next 200k
          pctincrease 0 maxextents 50)
          tablespace bd_data;
Index created.
```

2) Crearea unui index de tip BITMAP:

```
SQL>  create  bitmap index scott.dept_name_idx
          on scott.dept(dname)
          pctfree 30
          storage(initial 200k next 200k
          pctincrease 0 maxextents 50)
          tablespace   bd_data;
Index created.
```

3) Alocarea unei extensii pentru un index de tip B-Tree:

```
SQL> alter tablespace users
          add datafile 'e:/temp/index01.dbf'  size 10M;
Tablespace altered.

SQL> alter index emp_name_idx
          allocate extent (size 200k datafile ’e:/temp/index01.dbf’)
Index altered.

SQL> alter tablespace users drop datafile 'e:/temp/index01.dbf';
ERROR at line 1:
ORA-03262: the file is non-empty
```

Obs. Dupa ce se aloca o extensie unui index fisierul respectiv nu se poate sterge. Trebuie sters mai intai indexul.

4) Eliberarea spatiului neutilizat pentru un index de tip B-Tree:

```
SQL> alter index emp_name_idx
         deallocate unused;
Index altered.
```

5) Mutarea unui index in alt tablespace:

SQL> alter index emp_name_idx rebuild tablespace system;
Index altered.

6) Defragmentarea blocurilor unui index:

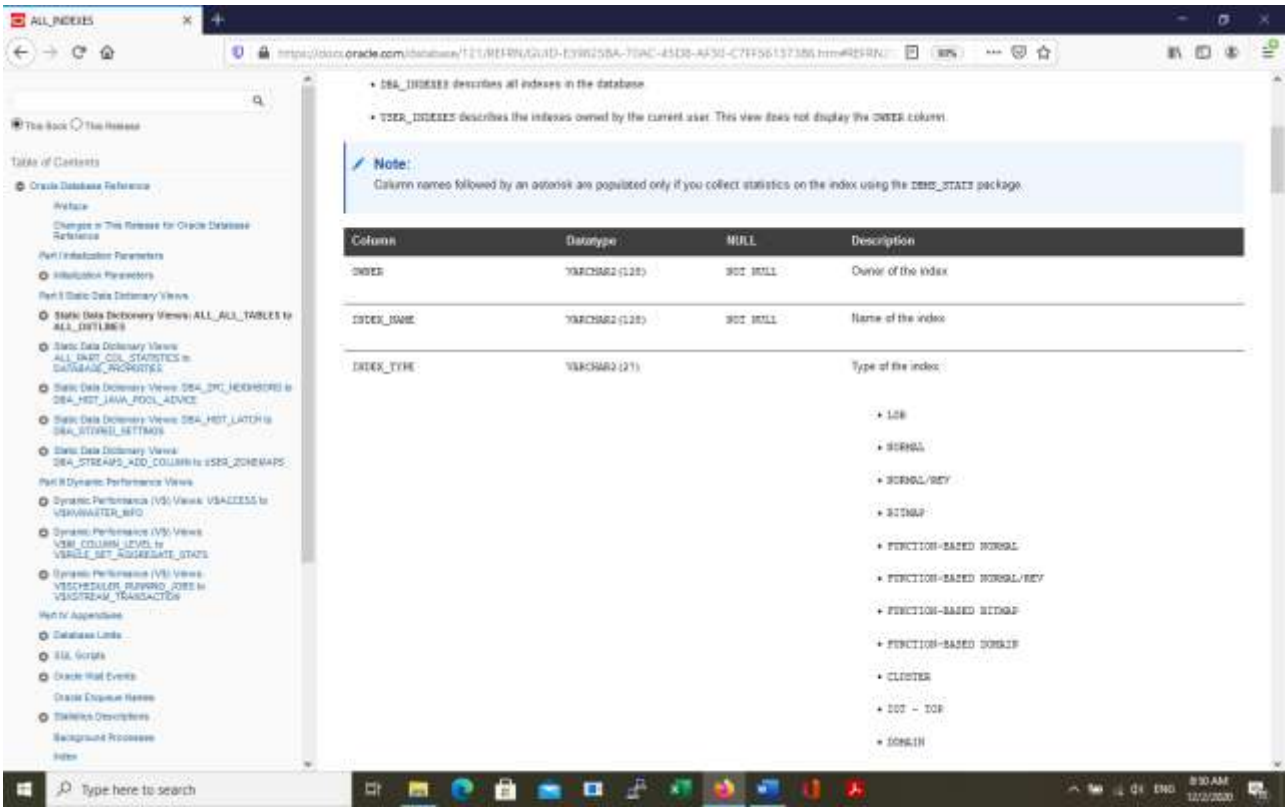
SQL> alter index emp_name_idx coalesce;
Index altered.

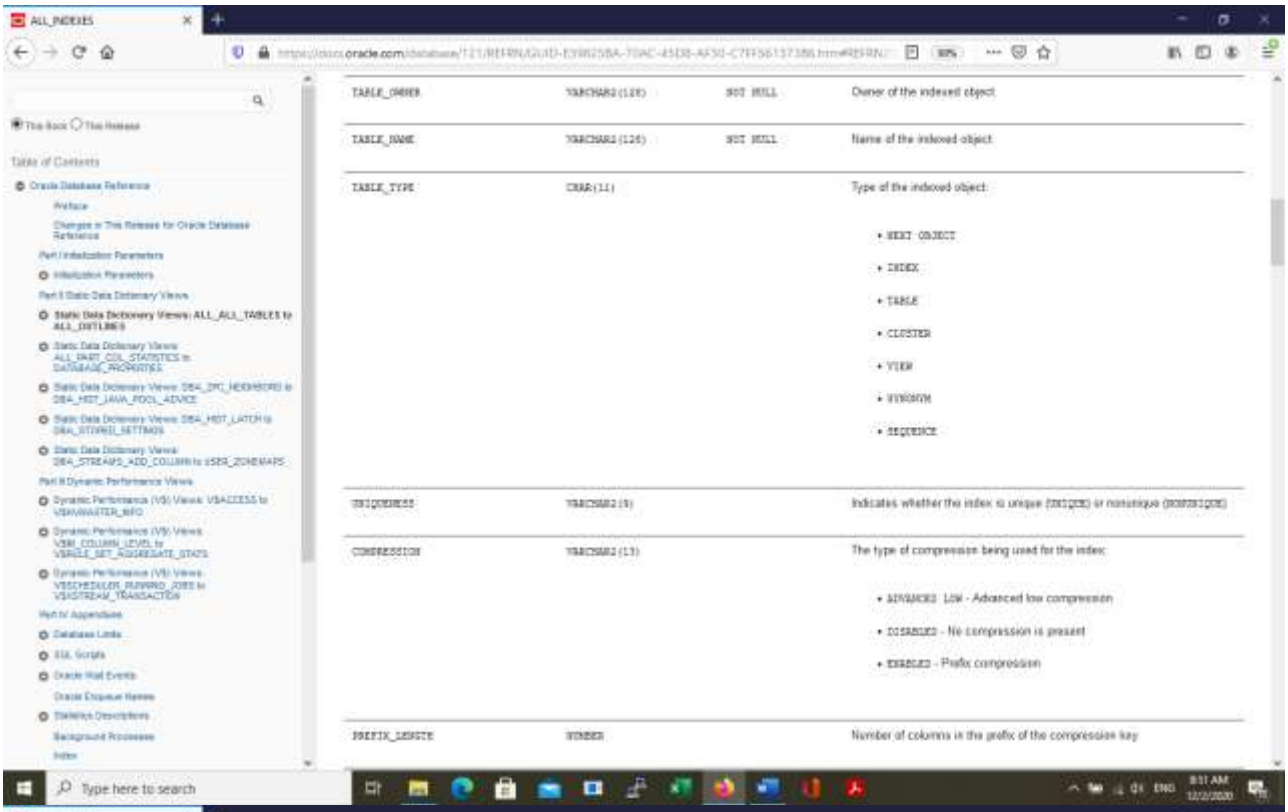
7) Informatii din dictionar despre indecsi:

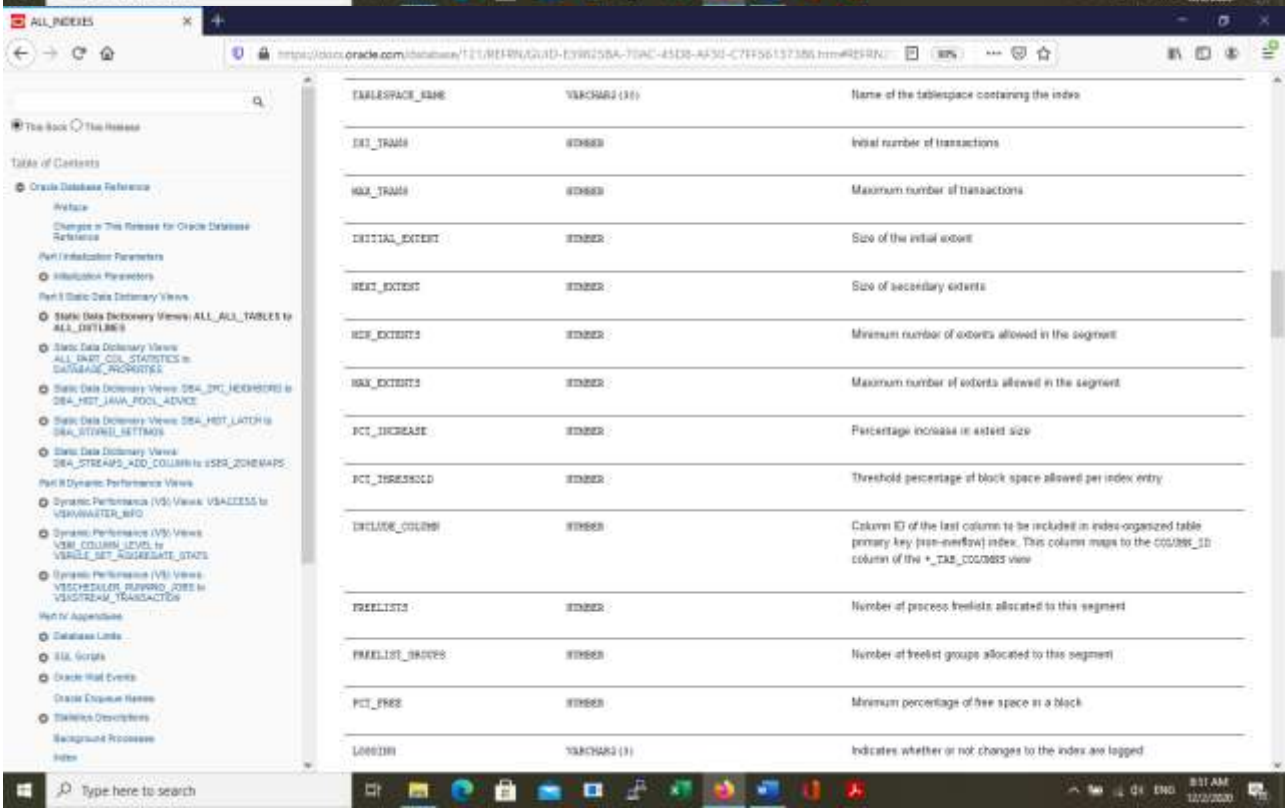
SQL> desc dba_indexes

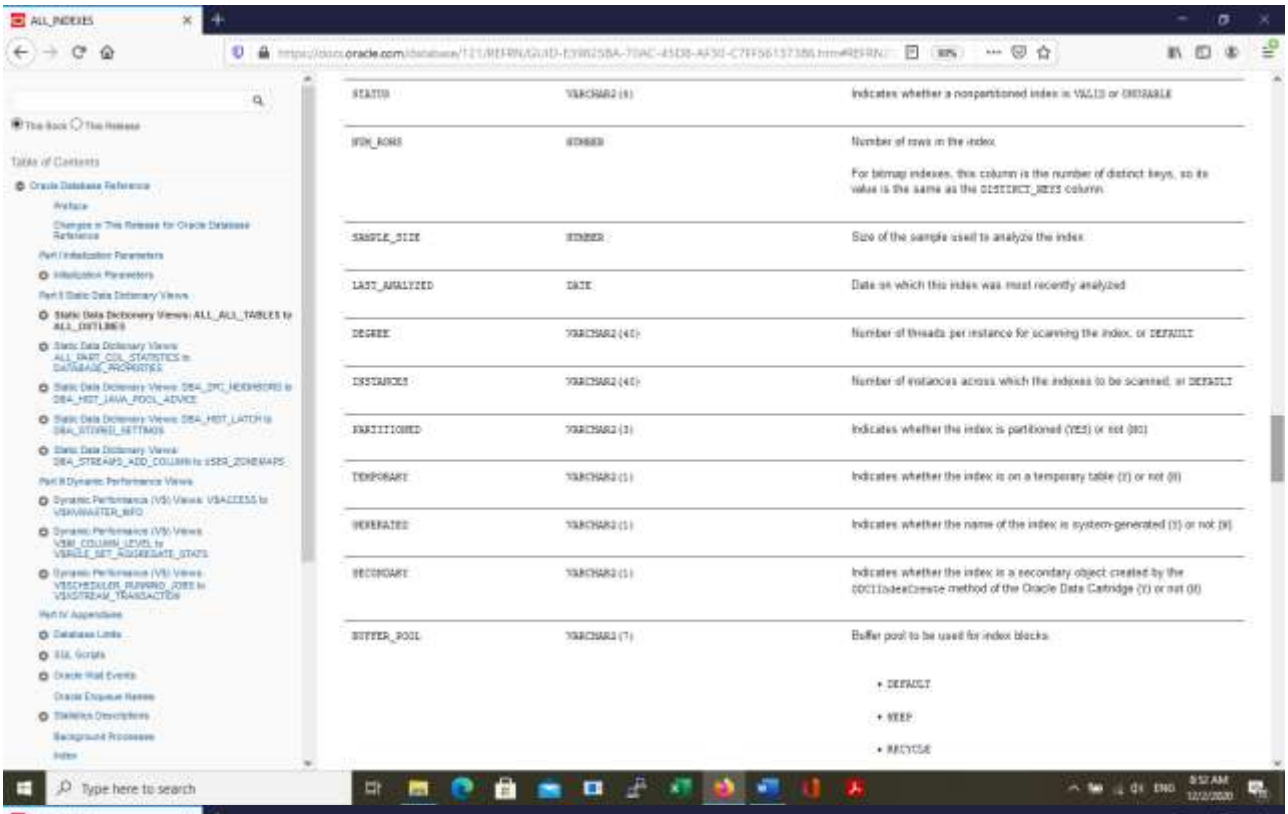
Name	Null?	Type
OWNER	NOT NULL	VARCHAR2(30)
INDEX_NAME	NOT NULL	VARCHAR2(30)
INDEX_TYPE		VARCHAR2(27)
TABLE_OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
TABLE_TYPE		VARCHAR2(11)
UNIQUENESS		VARCHAR2(9)
COMPRESSION		VARCHAR2(8)
PREFIX_LENGTH		NUMBER
TABLESPACE_NAME		VARCHAR2(30)
INI_TRANS		NUMBER
MAX_TRANS		NUMBER
INITIAL_EXTENT		NUMBER
NEXT_EXTENT		NUMBER
MIN_EXTENTS		NUMBER
MAX_EXTENTS		NUMBER
PCT_INCREASE		NUMBER
PCT_THRESHOLD		NUMBER
INCLUDE_COLUMN		NUMBER
FREELISTS		NUMBER
FREELIST_GROUPS		NUMBER
PCT_FREE		NUMBER
LOGGING		VARCHAR2(3)
BLEVEL		NUMBER
LEAF_BLOCKS		NUMBER
DISTINCT_KEYS		NUMBER
AVG_LEAF_BLOCKS_PER_KEY		NUMBER
AVG_DATA_BLOCKS_PER_KEY		NUMBER
CLUSTERING_FACTOR		NUMBER
STATUS		VARCHAR2(8)

NUM_ROWS	NUMBER
SAMPLE_SIZE	NUMBER
LAST_ANALYZED	DATE
DEGREE	VARCHAR2(40)
INSTANCES	VARCHAR2(40)
PARTITIONED	VARCHAR2(3)
TEMPORARY	VARCHAR2(1)
GENERATED	VARCHAR2(1)
SECONDARY	VARCHAR2(1)
BUFFER_POOL	VARCHAR2(7)
USER_STATS	VARCHAR2(3)
DURATION	VARCHAR2(15)
PCT_DIRECT_ACCESS	NUMBER
ITYP_OWNER	VARCHAR2(30)
ITYP_NAME	VARCHAR2(30)
PARAMETERS	VARCHAR2(1000)
GLOBAL_STATS	VARCHAR2(3)
DOMIDX_STATUS	VARCHAR2(12)
DOMIDX_OPSTATUS	VARCHAR2(6)
FUNCIDX_STATUS	VARCHAR2(8)
JOIN_INDEX	VARCHAR2(3)




ALL_INDEXES
TABLE_OWNER VARCHAR2 (128) NOT NULL Owner of the indexed object
TABLE_NAME VARCHAR2 (128) NOT NULL Name of the indexed object
TABLE_TYPE CHAR (11) Type of the indexed object:
• NEXT OBJECT
• INDEX
• TABLE
• CLUSTER
• VIEW
• SYNONYM
• SEQUENCE
UNIQUENESS VARCHAR2 (9) Indicates whether the index is unique (UNIQUE) or nonunique (NONUNIQUE)
COMPRESSION VARCHAR2 (13) The type of compression being used for the index:
• ADVANCED LOW - Advanced low compression
• DISABLED - No compression is present
• ENABLED - Prefix compression
PREFIX_LENGTH NUMBER Number of columns in the prefix of the compression key
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ALL_INDEXES
TABLESPACE_NAME VARCHAR2 (30) Name of the tablespace containing the index
INI_TRANS NUMBER Initial number of transactions
MAX_TRANS NUMBER Maximum number of transactions
INITIAL_EXTENT NUMBER Size of the initial extent
NEXT_EXTENT NUMBER Size of secondary extents
MIN_EXTENTS NUMBER Minimum number of extents allowed in the segment
MAX_EXTENTS NUMBER Maximum number of extents allowed in the segment
PCT_INCREASE NUMBER Percentage increase in extent size
PCT_THRESHOLD NUMBER Threshold percentage of block space allowed per index entry
INCLUDE_COLUMN NUMBER Column ID of the last column to be included in index-organized table primary key (non-overflow) index. This column maps to the COLUMN_ID column of the *_TAB_COLUMNS view
FREELISTS NUMBER Number of process freelists allocated to this segment
FREELIST_GROUPS NUMBER Number of freelist groups allocated to this segment
PCT_FREE NUMBER Minimum percentage of free space in a block
LOGGING VARCHAR2 (3) Indicates whether or not changes to the index are logged
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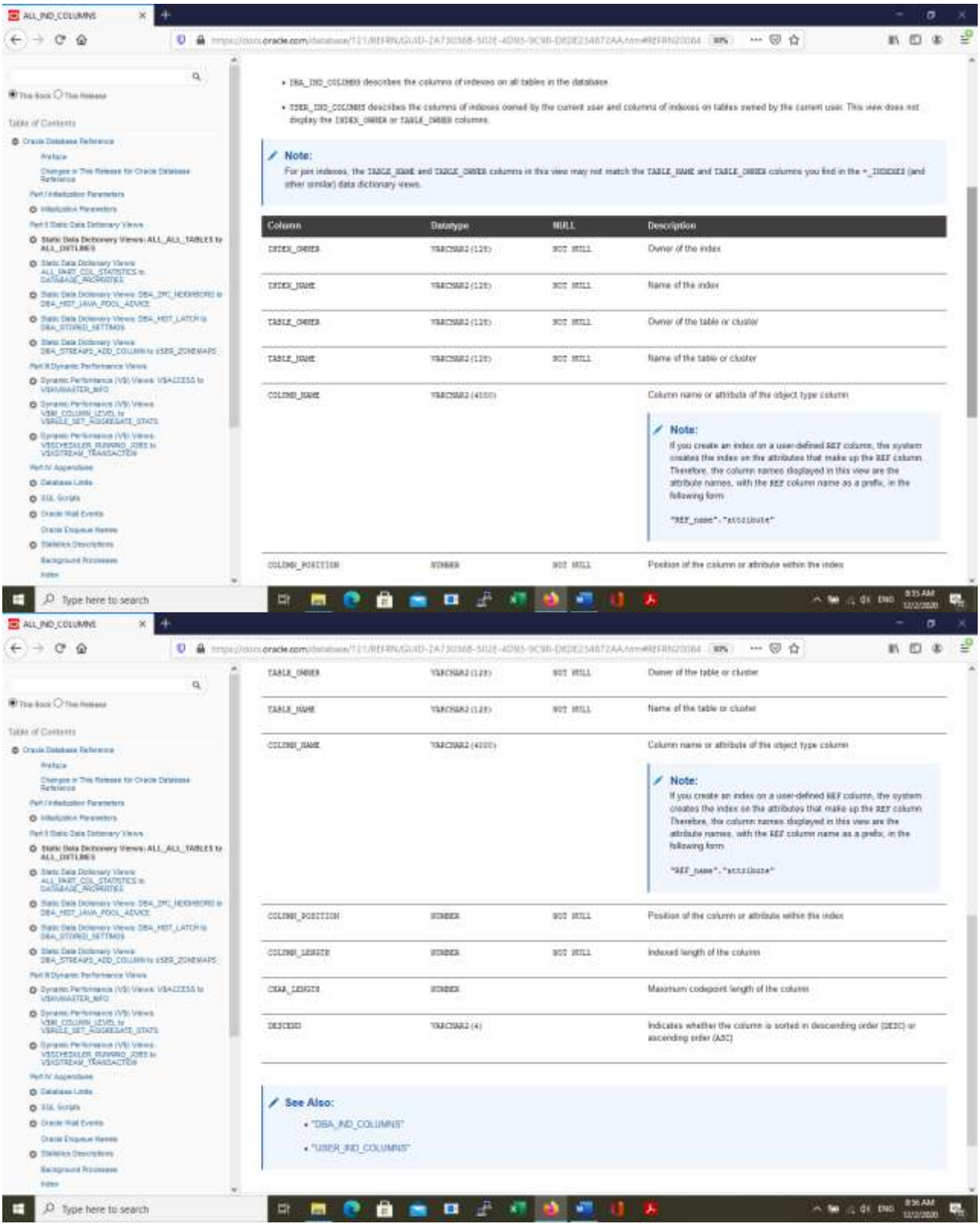

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STATUS VARCHAR2 (8) Indicates whether a nonpartitioned index is VALID or UNUSABLE
NUM_ROWS NUMBER Number of rows in the index
For bitmap indexes, this column is the number of distinct keys, so its value is the same as the DISTINCT_KEYS column.
SAMPLE_SIZE NUMBER Size of the sample used to analyze the index
LAST_ANALYZED DATE Date on which this index was most recently analyzed
DEGREE VARCHAR2 (40) Number of threads per instance for scanning the index, or DEFAULT
INSTANCES VARCHAR2 (40) Number of instances across which the indexes to be scanned, or DEFAULT
PARTITIONED VARCHAR2 (3) Indicates whether the index is partitioned (YES) or not (NO)
TEMPORARY VARCHAR2 (1) Indicates whether the index is on a temporary table (Y) or not (N)
GENERATED VARCHAR2 (1) Indicates whether the name of the index is system-generated (Y) or not (N)
SECONDARY VARCHAR2 (1) Indicates whether the index is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
BUFFER_POOL VARCHAR2 (7) Buffer pool to be used for index blocks
DEFAULT
KEEP
RECYCLE
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ENG
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ALL_INDEXES
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Index
IOT_REDUNDANT_PKEY_ELIM VARCHAR2 (3) Indicates whether redundant primary key columns are eliminated from secondary indexes on index-organized tables (YES) or not (NO)
DROPPED VARCHAR2 (3) Indicates whether the index has been dropped and is in the recycle bin (YES) or not (NO); NULL for partitioned tables
This view does not return the names of indexes that have been dropped.
VISIBILITY VARCHAR2 (9) Indicates whether the index is VISIBLE or INVISIBLE to the optimizer
DOMIDX_MANAGEMENT VARCHAR2 (14) If this is a domain index, indicates whether the domain index is system-managed (SYSTEM_MANAGED) or user-managed (USER_MANAGED)
SEGMENT_CREATED VARCHAR2 (3) Indicates whether the index segment has been created (YES) or not (NO)
ORPHANED_ENTRIES VARCHAR2 (3) Indicates whether a global index contains stale entries because of deferred index maintenance during DROP/TRUNCATE PARTITION or MODIFY PARTITION INDEXING OFF operations
Possible values:
YES - The index contains orphaned entries
NO - The index does not contain orphaned entries
INDEXING VARCHAR2 (7) Indicates whether a global index is decoupled from the underlying table
Possible values:
PARTIAL - The index is partial, that is, it will follow the table's indexing property
FULL - The index will include all partitions of the table
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```
SQL>  select  index_name, index_type, table_name, status from dba_indexes
          where owner='SCOTT' ;

INDEX_NAME                     INDEX_TYPE                 TABLE_NAME                     STATUS
------------------------------ -------------------------- ------------------------------ -------
DECIZII_PRIM                   NORMAL                     DECIZII                        VALID
DEPT_NAME_IDX                  BITMAP                     DEPT                           VALID
EMP_NAME_IDX                   NORMAL                     EMP                            VALID
PK_DEPT                        NORMAL                     DEPT                           VALID
PK_EMP                         NORMAL                     EMP                            VALID
PK_FUN                         NORMAL                     FUNCTII1                       VALID
PK_INT                         NORMAL                     INTRARI_GESTIUNE               VALID
PK_STOC                        NORMAL                     STOCURI                        VALID
```

8) Informatii din dictionar despre coloanele indecsilor:

```
SQL> desc dba_ind_columns
 Name                           Null?                  Type
 ------------------------------ --------               ----
 INDEX_OWNER                    NOT NULL               VARCHAR2(30)
 INDEX_NAME                     NOT NULL               VARCHAR2(30)
 TABLE_OWNER                    NOT NULL               VARCHAR2(30)
 TABLE_NAME                     NOT NULL               VARCHAR2(30)
 COLUMN_NAME                                           VARCHAR2(4000)
 COLUMN_POSITION                NOT NULL               NUMBER
 COLUMN_LENGTH                  NOT NULL               NUMBER
 CHAR_LENGTH                                           NUMBER
 DESCEND                                               VARCHAR2(4)
```


ALL_IND_COLUMNS
DBA_IND_COLUMNS describes the columns of indexes on all tables in the database.
USER_IND_COLUMNS describes the columns of indexes owned by the current user and columns of indexes on tables owned by the current user. This view does not display the INDEX_OWNER or TABLE_OWNER columns.
Note:
For join indexes, the TABLE_NAME and TABLE_OWNER columns in this view may not match the TABLE_NAME and TABLE_OWNER columns you find in the *_INDEXES (and other similar) data dictionary views.
Column
Datatype
NULL
Description
INDEX_OWNER
VARCHAR2 (128)
NOT NULL
Owner of the index
INDEX_NAME
VARCHAR2 (128)
NOT NULL
Name of the index
TABLE_OWNER
VARCHAR2 (128)
NOT NULL
Owner of the table or cluster
TABLE_NAME
VARCHAR2 (128)
NOT NULL
Name of the table or cluster
COLUMN_NAME
VARCHAR2 (4000)
Column name or attribute of the object type column
Note:
If you create an index on a user-defined REF column, the system creates the index on the attributes that make up the REF column. Therefore, the column names displayed in this view are the attribute names, with the REF column name as a prefix, in the following form:
"REF_name"."attribute"
COLUMN_POSITION
NUMBER
NOT NULL
Position of the column or attribute within the index
Type here to search
ALL_IND_COLUMNS
TABLE_OWNER
VARCHAR2 (128)
NOT NULL
Owner of the table or cluster
TABLE_NAME
VARCHAR2 (128)
NOT NULL
Name of the table or cluster
COLUMN_NAME
VARCHAR2 (4000)
Column name or attribute of the object type column
Note:
If you create an index on a user-defined REF column, the system creates the index on the attributes that make up the REF column. Therefore, the column names displayed in this view are the attribute names, with the REF column name as a prefix, in the following form:
"REF_name"."attribute"
COLUMN_POSITION
NUMBER
NOT NULL
Position of the column or attribute within the index
COLUMN_LENGTH
NUMBER
NOT NULL
Indexed length of the column
CHAR_LENGTH
NUMBER
Maximum codepoint length of the column
DESCEND
VARCHAR2 (4)
Indicates whether the column is sorted in descending order (DESC) or ascending order (ASC)
See Also:
"DBA_IND_COLUMNS"
"USER_IND_COLUMNS"
Type here to search

```
SQL> select  index_name, table_owner, table_name, column_name
        from dba_ind_columns  where index_owner='SCOTT';

INDEX_NAME                     TABLE_OWNER                    TABLE_NAME                     COLUMN_NAME
------------------------------ ------------------------------ - ------------------------------ ----------------
DEPT_NAME_IDX                  SCOTT                          DEPT                             DNAME
EMPNAME_IDX                    SCOTT                          EMP                              ENAME
PK_COMP                        SCOTT                          COMPONENTE                       COD_COMP
PK_COMP                        SCOTT                          COMPONENTE                       PRET
PK_DEPT                        SCOTT                          DEPT                             DEPTNO
PK_EMP                         SCOTT                          EMP                              EMPNO
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 NR_DOC_IN
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 DATA_DOC_IN
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 COD_PRODUS
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 COD_UM
PK_STOC                        SCOTT                          STOCURI                          COD_COMP
PK_STOC                        SCOTT                          STOCURI                          PRET
PK_STOC                        SCOTT                          STOCURI                          DATA_STOC
```

9) Startarea si stoparea monitorizarii unui index:

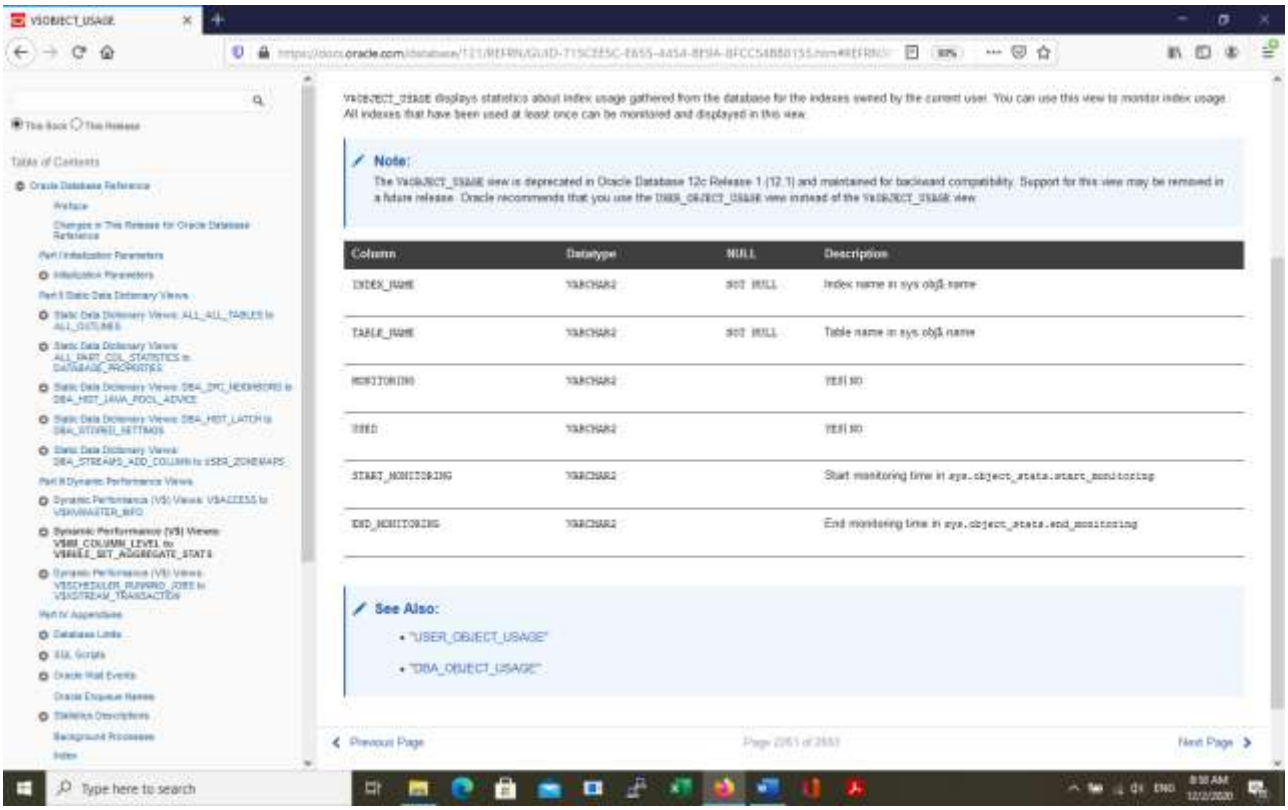
```
SQL> alter index emp_name_idx
        monitoring usage;
Index altered.

SQL> alter index emp_name_idx
        nomonitoring usage;
Index altered.
```

10) Informatii din dictionar despre indecsii monitorizati:

```
SQL> desc v$object_usage   ( sau  user_object_usage)

 Name                           Null?            Type
 ------------------------------ --------         ----
 INDEX_NAME                      NOT NULL        VARCHAR2(30)
 TABLE_NAME                      NOT NULL        VARCHAR2(30)
 MONITORING                                      VARCHAR2(3)
 USED                                            VARCHAR2(3)
 START_MONITORING                                 VARCHAR2(19)
 END_MONITORING                                   VARCHAR2(19)
```



SQL> select * from v$object_usage;

INDEX_NAME	TABLE_NAME	MON	USE	START_MONITORING	END_MONITORING
EMP_NAME_IDX	EMP	NO	NO	11/15/2019 15:38:30	11/15/2019 15:41:26

11) Startarea analizei structurii unui index (se populeaza view-ul INDEX_STATS cu informatii despre index):

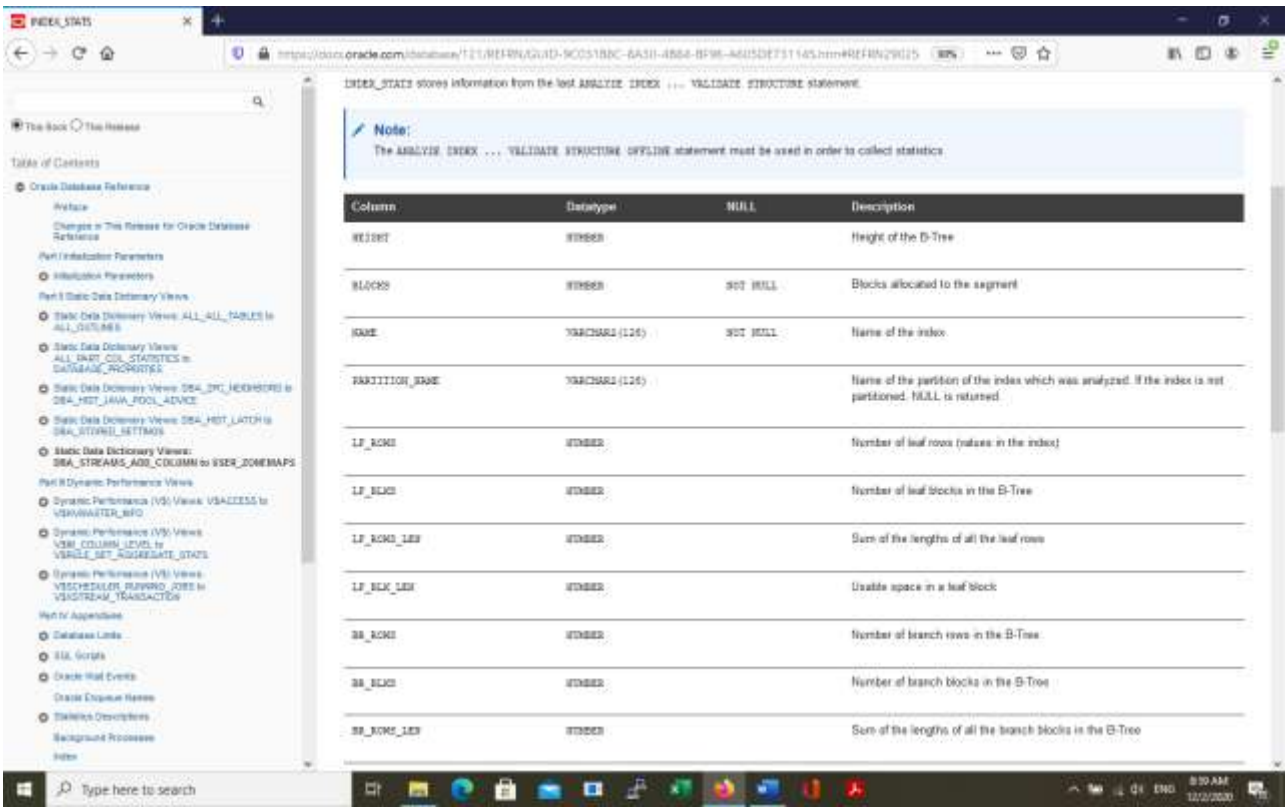
SQL> analyze index emp_name_idx validate structure;
Index analyzed.

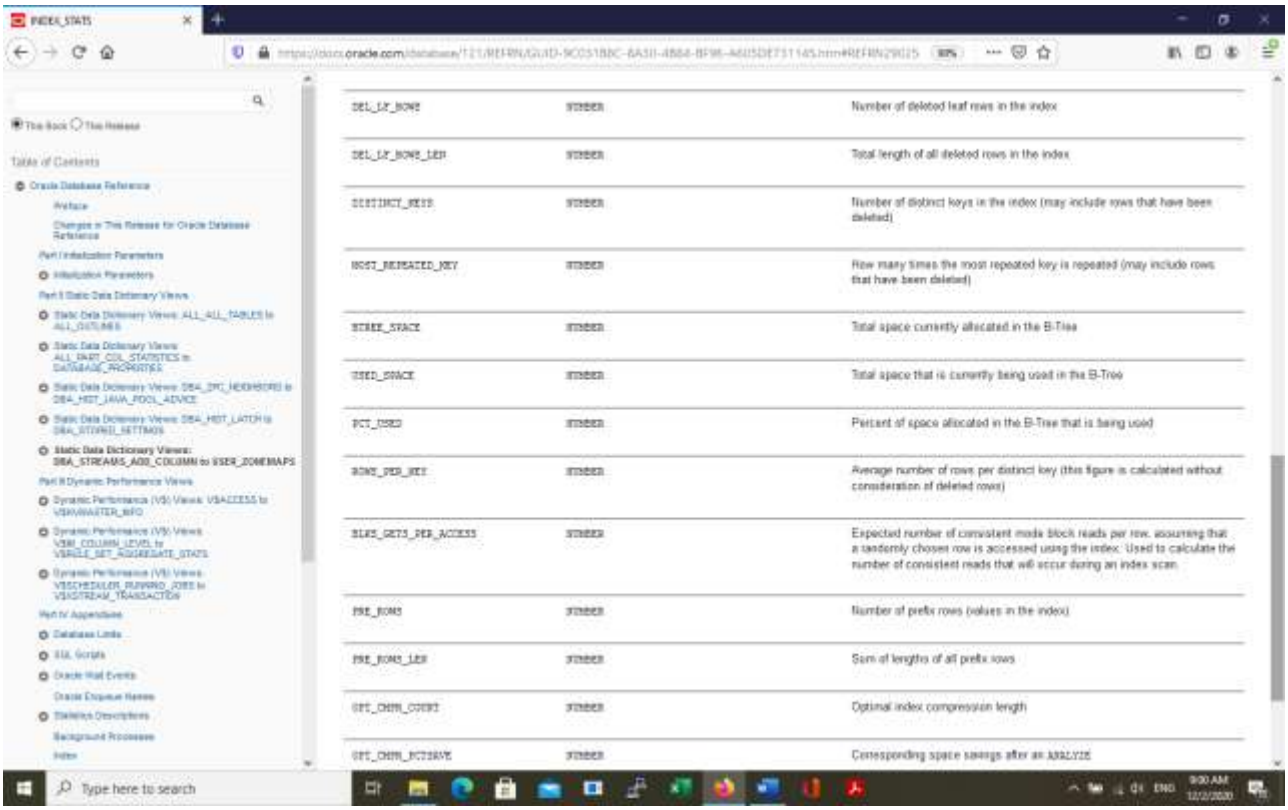
12) Informatii din dictionar despre starea indecsilor:

SQL> desc index_stats

Name	Null?	Type
HEIGHT		NUMBER
BLOCKS		NUMBER
NAME		VARCHAR2(30)
PARTITION_NAME		VARCHAR2(30)
LF_ROWS		NUMBER
LF_BLKS		NUMBER

LF_ROWS_LEN	NUMBER
LF_BLK_LEN	NUMBER
BR_ROWS	NUMBER
BR_BLKS	NUMBER
BR_ROWS_LEN	NUMBER
BR_BLK_LEN	NUMBER
DEL_LF_ROWS	NUMBER
DEL_LF_ROWS_LEN	NUMBER
DISTINCT_KEYS	NUMBER
MOST_REPEATED_KEY	NUMBER
BTREE_SPACE	NUMBER
USED_SPACE	NUMBER
PCT_USED	NUMBER
ROWS_PER_KEY	NUMBER
BLKS_GETS_PER_ACCESS	NUMBER
PRE_ROWS	NUMBER
PRE_ROWS_LEN	NUMBER
OPT_CMPR_COUNT	NUMBER
OPT_CMPR_PCTSAVE	NUMBER





SQL> select name, blocks, used_space, pct_used, distinct_keys, lf_rows, del_lf_rows
from index_stats ;

NAME	BLOCKS	USED_SPACE	PCT_USED	DISTINCT_KEYS	LF_ROWS	DEL_LF_ROWS
EMP_NAME_IDX	32	409	6	23	23	0

13) Stergerea unui index din dictionar:

SQL> drop index emp_name_idx ;
Index dropped.

Exercitii :

1. Sa se creeze o copie a tabelei EMP apoi sa se creeze un index pe coloanele deptno si empno din tabela copie. Verificati in dictionar componenta indexului.

2. Verificati in dictionar numarul de blocuri alocate indexului si care este procentul utilizat din spatiul alocat.

3. Faceti o lista cu numele instantei curente, numele indecsilor si tabelele aferente.

4. Monitorizati indexul creat si verificati in dictionar data si ora cand a inceput monitorizarea .

5. Sa se creeze un nou tablespace TAB_IDX intr-un fisier TAB_IDX_FIS cu dimensiunea de 2 Mb. Sa se mute apoi indexul creat anterior in noul tablespace. La sfarsit se sterg din dictionar indexul si tablespace-ul.